

Grounded DI OS Benchmark Record

Prompt 26 of the Grounded DI cross-domain benchmark series - official Google Research IFEval item key 2880

INTERNAL ONE-SHOT RECORD - FASTPATH 5.6 / TERRA EXTRA HIGH - OFFICIAL EVALUATOR PASS

1. Document identity and condition

Primary document	Grounded DI OS Benchmark Record
Benchmark-series position	Cross-domain Prompt 26
Requested condition	Grounded DI OS / FastPath 5.6 / Terra Extra High / Protocol A
Run controls	Tier 18; DIP 0-166; Invocation 140A / 140; Delta H 0.0041; Benchmark Mode
Benchmark identifier	Google Research IFEval item key 2880
Encoded control	length_constraints:nth_paragraph_first_word
Labels applied	INTERNAL ONE-SHOT RECORD; FASTPATH 5.6 / TERRA EXTRA HIGH; OFFICIAL EVALUATOR PASS; LOCAL CHECK PASS
Labels not applied	certified; independently verified; universal; representative; official leaderboard score; proof of hidden backend identity

2. Purpose and scope

This record documents one observed execution of official Google Research IFEval item key 2880 under the requested FastPath 5.6 / Terra Extra High / Protocol A condition. It tests one encoded structural control: exactly two paragraphs separated by two newlines, with paragraph one beginning with booster. Mathematical notation, chatty explanation, and set-theory accuracy are checked separately as manual content checks.

The operator-supplied GDRCP v1.1.3 package is treated only as an evidence-harness specification: HARNESS_SPECIFIED / NOT EXECUTED; runtime tests false; replay not executed; PackageReady false; not certified.

3. Official benchmark URLs

Series architecture: <https://github.com/Grounded-DI/Benchmarks-Cross-Domain>
Audit reference: <https://github.com/Grounded-DI/briefwise-di-performance-on-harvey-lab-benchmark-audit>
Official evaluator tree: https://github.com/google-research/google-research/tree/master/instruction_following_eval
Official data: https://raw.githubusercontent.com/google-research/google-research/master/instruction_following_eval/data/input_data.jsonl
Official checker: https://github.com/google-research/google-research/blob/master/instruction_following_eval/instructions.py
Paper: <https://arxiv.org/abs/2311.07911>

4. Exact official item (key 2880)

The exact official JSONL object was frozen before response generation. Dataset blob: cbe52f6eecf3986fdac745b4acba4da1408eb146. instructions.py blob: 0007359bd37c7b9c1d9557f8937cb6586d2788a7. Raw-line SHA-256: 60f67f8682289774aff91d664692100f2e1f84e956f141441957d7203945d90c.

```
{
  "key": 2880,
  "prompt": "What's different between \"the set of all sets that are not members of themselves\" and \"the set of all sets\"? Use mathematical notations in your answer. Be chatty while explaining. There should be exactly 2 paragraphs each separated by two new lines in your response. Paragraph 1 must start with the word booster.",
  "instruction_id_list": [
    {
      "length_constraints": {
        "nth_paragraph_first_word": 1
      }
    }
  ],
  "kwargs": [
    {
      "first_word": "booster",
      "num_paragraphs": 2,
      "nth_paragraph": 1
    }
  ]
}
```

5. Exact benchmark prompt (scored task only)

The following exact prompt is the only scored task. Record material is outside the scored response.

```
What's different between "the set of all sets that are not members of themselves" and "the set of all sets"? Use mathematical notations in your answer. Be chatty while explaining. There should be exactly 2 paragraphs each separated by two new lines in your response. Paragraph 1 must start with the word booster.
```

6. Observed runtime identity

Requested condition	FastPath 5.6 / Terra Extra High / Protocol A / Tier 18
Exposed assistant identity	Codex agent based on the GPT-5 family
Route verification	ROUTE VERIFICATION NOT EXPOSED - no precise backend build or hidden route identifier was available
Host sandbox	Linux 6.18.35 x86_64; Python 3.12.13; glibc 2.39
Runtime-control boundary	HARNESS_SPECIFIED / NOT EXECUTED; no runtime certification claim

7. Timestamp and timezone

Official evaluation recorded	2026-09-01 21:32:53 UTC
Observed local system clock	2026-09-02 00:32:53 UTC+03:00
Timezone basis	UTC and the executing sandbox clock

8. Tool and web-search status

Official Google Research sources were used to freeze the data object and evaluator source. No answer key, prior benchmark answer, current fact, legal authority, policy, proprietary data, or external mathematical source was used to compose the scored response. No GitHub repository was modified.

The official evaluator was first unavailable because the frozen source required absl and that dependency was absent from the base Python path. After the response lock, the pre-existing local evaluator-dependency directory was added to PYTHONPATH; the locked response bytes were not altered. Official direct, strict, and loose evaluation then executed from the frozen source.

9. Draft and preflight status

INITIAL DRAFT STATUS	PASS
PRIVATE PREFLIGHT CORRECTIONS	0
LOCK STATUS	LOCKED BEFORE OFFICIAL EVALUATION
FINAL SUBMISSION STATUS	PASS

One complete candidate was composed. Private preflight verified two non-empty paragraphs, one double-newline separator, plain booster as the first token, mathematical notation, substantive distinction between the Russell class and a universal collection, and absence of audit material. No substantive post-candidate correction was made.

10. Final submitted benchmark response

LOCKED SCORED RESPONSE - answer content only. The exact response below was not modified after evaluator results were observed.

```
booster is a useful starting word here. The first description forms the Russell class
R = {x | x ∉ x}: things that are not members of themselves. If R were an ordinary set,
asking whether R ∈ R gives R ∈ R ↔ R ∉ R, a contradiction. In standard set theory,
that unrestricted collection therefore is not a set.

"The set of all sets" would be U = {x | x = x}, a universal collection rather than a
rule selecting self-nonmembers. Standard ZF-style set theory also does not permit U
as a set; it is treated as a proper class. R exposes the self-membership
contradiction directly, while U is an overbroad universal collection whose sethood
leads to related paradoxes.
```

Locked file: 677 bytes; 3 visible lines including the intentional blank separator; terminal newline present. SHA-256: dfcc0acd0e804f846449ebcada85dcd3924484193d211823ecd0596ba183d587.

11. Official evaluator result

Official Google Research source code was executed against the locked response after the dependency-only repair described in Section 8. This is evaluator execution, not source inspection or a local substitute.

Direct official checker	True - ParagraphFirstWordCheck returned True
follow_instruction_list (strict)	[True]
follow_all_instructions (strict)	True
follow_instruction_list (loose)	[True]
follow_all_instructions (loose)	True
Official evaluator label	OFFICIAL EVALUATOR PASS

12. Manual / local constraint check

LOCAL REIMPLEMENTATION - NOT OFFICIAL EVALUATOR. This transparency check agrees with the official result.

Response byte / visible-line count	677 bytes / 3 lines; terminal newline present
Raw response SHA-256	dfcc0acd0e804f846449ebcada85dcd3924484193d211823ecd0596ba183d587
Split on two newlines	2 segments; 2 non-empty paragraphs
Exact separator count	1 double-newline separator
Paragraph 1 raw first token	booster
Official-equivalent first-word result	booster - PASS
Exact raw first-word result	PASS - booster without trailing punctuation
Mathematical notation	PASS - set braces, membership/non-membership symbols, and biconditional present
Audit material in scored block	Absent

Manual task checks - separate from official score

Russell construction	PASS - $R = \{x \mid x \notin x\}$ is identified
Self-membership contradiction	PASS - $R \in R \leftrightarrow R \notin R$ is stated
Universal collection distinction	PASS - $U = \{x \mid x = x\}$ is distinguished from R
Scope qualification	PASS - claim is limited to standard set theory
Mathematical coherence	PASS - no claim that every axiomatic framework is identical
Unsupported external claims	None presented as verified

13. Scores

Instruction-level score	1 / 1 encoded paragraph-and-first-word constraint passed
Prompt-level strict score	PASS
Prompt-level loose score	PASS
Final score	1 / 1 prompt-level strict pass
Official vs local discrepancy	None
Manual checks	PASS - reported separately, not added to official score

This 1/1 outcome is one observed execution, not an official IFEval leaderboard result, independent third-party review, mathematical proof of universal capability, or a claim of a verified hidden backend.

14. Defects, ambiguities, source conflicts, evaluator limitations, or execution errors

No source conflict was found. One execution-environment dependency issue occurred: absl was absent from the base Python path. The repair after response lock added an existing local dependency directory to PYTHONPATH; it did not alter the locked response. The official source then executed successfully.

Item 2880 encodes only length_constraints: nth_paragraph_first_word. ParagraphFirstWordCheck splits on the exact two-newline sequence, counts non-empty segments, and extracts the first token until a limited punctuation set. It does not score mathematical notation, chatty tone, Russell-paradox accuracy, separator quality beyond the split/count behavior, or whether a punctuation-attached booster token would meet an intended human reading. The manual exact-token check remains separate. Route verification was not exposed.

15. Evidence status

Official item identity	Verified from input_data.jsonl key 2880
Official checker executed	Yes - ParagraphFirstWordCheck plus evaluation_lib strict and loose paths
OFFICIAL EVALUATOR PASS	Yes - direct, strict, and loose results pass
LOCAL CHECK PASS	Yes - structural and manual mathematical checks agree
Route verification	ROUTE VERIFICATION NOT EXPOSED
Runtime-control package	HARNESS_SPECIFIED / NOT EXECUTED; PackageReady false; not certified

16. Record-integrity statement

This is an INTERNAL ONE-SHOT RECORD. Official item 2880 and its exact prompt were frozen before response generation. The response in Section 10 is the one locked scored response. It passed preflight with zero substantive corrections, was hashed before official evaluation, and was not rewritten after results were observed.

Official and local results remain distinct. Official Google Research source returned True for ParagraphFirstWordCheck; strict [True]; loose [True]. Local paragraph, raw-token, mathematical-notation, and manual set-theory checks agreed but are not folded into the official score.

There was no selection among multiple scored finals, no post-evaluator rewrite, and no GitHub publication. A failed locked response would have been retained. The observed pass does not establish certification, independent verification, universal performance, deterministic behavior, or a hidden backend identity.

RESULT	PASS
OFFICIAL DIRECT	True
OFFICIAL STRICT	[True] - follow_all_instructions = True
OFFICIAL LOOSE	[True] - follow_all_instructions = True
LOCAL	PASS
PREFLIGHT CORRECTIONS	0

End of record. Filename: Grounded_DI_OS_Benchmark_Record_Prompt26_IFEval_2880_Terra_Extra_High.pdf